

Nguyen Huu Thang

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🐙 github.com/InfinityZero3000 | 🌐 [Portfolio Website](#) | 📍 Ho Chi Minh City, Vietnam

SUMMARY

Software Engineering student focused on NLP, LLM applications, and AI product engineering. Experienced in building Vietnamese NLP datasets, fine-tuning transformer models, designing knowledge-graph retrieval pipelines, and serving ML features through FastAPI services. Seeking an AI Engineer Internship to ship research-grade ML into practical software systems.

EDUCATION

Ho Chi Minh City University of Industry and Trade (HUIT)

Bachelor of Information Technology – Software Engineering

Sep. 2023 – Present

Ho Chi Minh City, Vietnam

TECHNICAL SKILLS

Programming & Backend: Python, C#, SQL, FastAPI, ASP.NET Core, REST APIs.

AI/Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LoRA/PEFT.

NLP & LLMs: Transformers, PhoBERT, Qwen, RAG, LangChain, LangGraph, Knowledge Graphs, Whisper, Piper.

Data, Mobile & Tools: PostgreSQL, MongoDB, Redis, KuzuDB, Docker, Git, Flutter.

EXPERIENCE

Web Collaborator

Sigma Group

Nov. 2025 – Jan. 2026

Ho Chi Minh City, Vietnam

- **Developed and customized** responsive production pages in Liquid for an insurance sales website, converting design mockups and business requirements into customer-facing interfaces.
- **Implemented** reusable page sections, content blocks, and layout adjustments across desktop and mobile breakpoints to improve interface consistency and usability.
- **Handled** iterative feedback from the project team and delivered targeted UI updates while preserving existing styles, component behavior, and page structure.

PROJECTS

ViRes-HotelABSA – Vietnamese Aspect-Based Sentiment Analysis

Paper

Research Project | Team of 3 | Python, PyTorch, PhoBERT, Knowledge Graphs, CRF

NLP / AI Research

- **Built** ViRes-HotelABSA, a Vietnamese ABSA corpus with 10,406 hotel reviews and 47,163 entity-aspect-sentiment triples using LLM-assisted annotation and human review.
- **Measured** annotation quality with Cohen's $\kappa = 0.88$ for aspect-boundary identification and $\kappa = 0.97$ for sentiment-polarity classification.
- **Developed** PhoBERT-KGAN-CRF by combining PhoBERT with a 143K-node, 6.38M-edge domain knowledge graph, BIOES sequence labeling, and CRF decoding.
- **Achieved** 77.30% F1 for aspect extraction and 81.28% F1 for sentiment classification, outperforming PhoBERT-based baselines under the same experimental setting.

LexiLingo – AI English Tutoring Platform

GitHub | Live Demo

Personal Project | Python, FastAPI, Flutter, PostgreSQL, Redis, Qwen, LoRA/PEFT

AI / Full-Stack Development

- **Built** LexiLingo, a 5-service AI English-tutoring platform with Flutter mobile, FastAPI backend, Python AI service, React admin dashboard, MCP tooling, PostgreSQL, and Redis.
- **Designed** TraceCAG, a LangGraph-based tutoring pipeline that retrieves learner profiles, lesson context, and KuzuDB knowledge-graph facts before generating grammar corrections, vocabulary hints, and tutor responses.
- **Integrated** Faster-Whisper STT and Piper TTS for two-way speaking practice, including speech recording, transcription, correction, and spoken feedback across 68 topic catalogs.
- **Fine-tuned** a 1.7B Qwen model with LoRA/PEFT for grammar correction and contextual vocabulary suggestions, then exposed inference through FastAPI endpoints.

HONORS & AWARDS

DASGRI Co-Author – Presented knowledge-graph-enhanced PhoBERT research for Vietnamese ABSA.

2026

1st Prize – HUIT Scientific Research – Developed ViRes-HotelABSA for Vietnamese ABSA.

2026

Data Science Finalist – Reached the final round and contributed to feature engineering and model evaluation.

2025

LANGUAGES

English: Technical reading and writing; conversational communication — **Vietnamese:** Native.